

COMP 324 Network Security

Module 3 Lab 2: File Operations and Permissions

Lab Report

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Exact Commands Run

Step 1: Creating Files and Directories

```
mkdir -p /home/student1/project
```

```
touch /home/student1/project/test.txt
```

```
echo "This is a test file for access control lab" > /home/student1/project/test.txt
```

```
ls -la /home/student1/project/
```

Step 2: Understanding File Permissions

```
ls -la /home/student1/project/test.txt
```

Step 3: Changing File Ownership

```
sudo chown root:root /home/student1/project/test.txt
```

```
ls -la /home/student1/project/test.txt
```

Step 4: Testing Access as Student

```
su - student1 -c "echo 'test' >> /home/student1/project/test.txt"
```

Step 1: Creating Files and Directories

```
[student1@threepio ~]$ mkdir -p /home/student1/project
[student1@threepio ~]$ touch /home/student1/project/test.txt
[student1@threepio ~]$ echo "This is a test file for access control lab" > /home/student1/project/test.txt
[student1@threepio ~]$ ls -la /home/student1/project/
total 12
drwx----- 2 student1 student1 4096 Sep 28 21:32 .
drwx----- 4 student1 student1 4096 Sep 28 21:32 ..
-rw-r--r-- 1 student1 student1  43 Sep 28 21:40 test.txt
[student1@threepio ~]$
```

Step 2: Understanding File Permissions

```
[student1@threepio ~]$ ls -la /home/student1/project/test.txt
-rw-r--r-- 1 student1 student1 43 Sep 28 21:47 /home/student1/project/test.txt
[student1@threepio ~]$
```

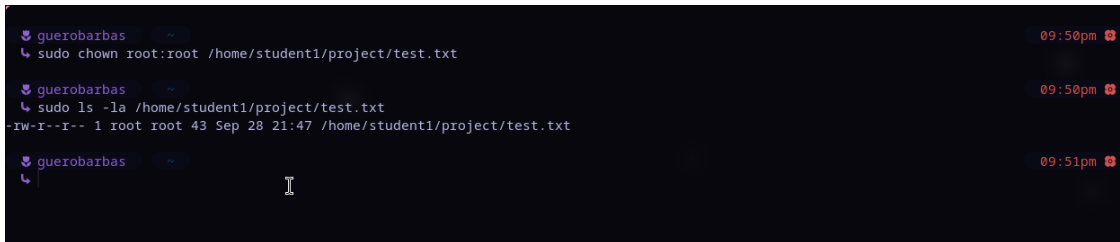
Recorded output:

```
[student1@threepio ~]$ ls -la /home/student1/project/test.txt
-rw-r--r-- 1 student1 student1 43 Sep 28 21:47 /home/student1/project/test.txt
```

Field Breakdown

- **File permissions:** `-rw-r--r--` (regular file, **owner:** read/write, **group:** read only, **others:** read only)
- **Hard link count:** 1 → only one file name points to this file.
- **User owner:** student1 → the user that owns the file.
- **Group owner:** student1 → the group with access permissions.
- **File size (bytes):** 43 → the file contains 43 bytes of data.
- **Last modified date/time:** Sep 28 21:47 → when the file was last changed.
- **Filename:** /home/student1/project/test.txt → the file's full path.

Step 3: Changing File Ownership

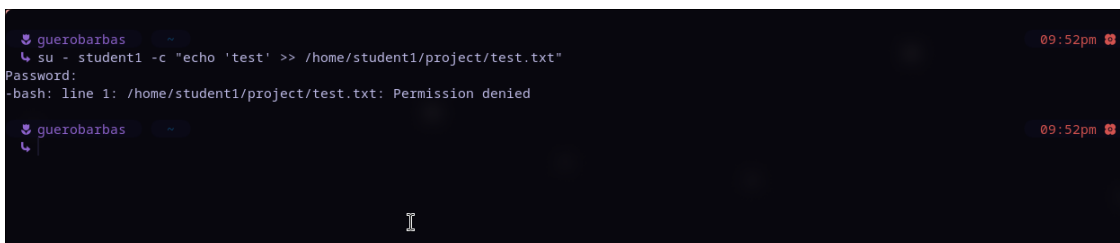


```
guero@barbas ~$ sudo chown root:root /home/student1/project/test.txt
guero@barbas ~$ sudo ls -la /home/student1/project/test.txt
-rw-r--r-- 1 root root 43 Sep 28 21:47 /home/student1/project/test.txt
```

Updated ownership:

```
-rw-r--r- 1 root root 43 Sep 28 21:47 /home/student1/project/test.txt
```

Step 4: Testing Access as Student



```
guero@barbas ~$ su - student1 -c "echo 'test' >> /home/student1/project/test.txt"
Password:
-bash: line 1: /home/student1/project/test.txt: Permission denied
```

Recorded error message:

```
-bash: line 1: /home/student1/project/test.txt: Permission denied
```

Error Explanation:

This occurred because the file is now owned by root, and student1 only has read access, not write access.

Observations:

- File ownership decides which user can read, write, or modify a file.
- Changing ownership with **chown** can remove the original user's ability to edit their own file.
- Permissions **rw-r-r-** allow only the owner to write, while group and others have read-only access.
- Attempting to append text as a non-owner is prohibited by Linux security.

Conclusion

File ownership and permissions are very important in Linux. This lab showed how creating files, changing ownership, and testing access shows how serious Linux is about its permissions. Only the owner or root can modify files with no restriction, while others are restricted.